Title of the Project

UCSC398J

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Science

in

Computer Science

by

Student NAME

REGISTER NUMBER

Under the guidance of

Prof. / Dr.

School of Computer Science Engineering and Information Systems

VIT, Vellore

April, 2026

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DECLARATION

I hereby declare that the UCSC398J – Project thesis entitled "**Thesis Title**" submitted by me, for the award of the degree of ***Bachelor of Science in Computer Science, School of Computer Science Engineering and Information Systems*** to VIT is a record of bonafide work carried out by me under the supervision **Guide Name, Designation, SCORE, VIT, Vellore.**

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree ordiploma in this institute or any other institute or university.

Place: Vellore

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CERTIFICATE

This is to certify that the UCSC398J – Project thesis entitled "**Thesis Title**" submitted by Student Name & Reg. No, SCORE, VIT, for the award of the degree of ***Bachelor of Science in Computer Science, School of Computer Science Engineering and Information Systems***, is a record of bonafide work carried out by him / her under my supervision during the period, 05. 12. 2025 to 17. 04. 2026, as per the VIT code of academic and research ethics.

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Place: Vellore

Date:

Signature of the Guide

Internal Examiner **External Examiner**

Head of the Department

Department of Computer Applications

ACKNOWLEDGEMENT

It is my pleasure to express with a deep sense of gratitude to my **UCSC398J - Project** guide **<Dr. or Prof. Guide Name, Designation>,** School of Computer Science Engineering and Information Systems, Vellore Institute of Technology, Vellore for **his/her** constant guidance, continual encouragement, in my endeavor. My association with **him/her** is not confined to academics only, but it is a great opportunity on my part to work with an intellectual and an expert in the field of **< Domain >.**

"I would like to express my heartfelt gratitude to Honorable Chancellor **Dr. G Viswanathan**; respected Vice Presidents **Mr. Sankar Viswanathan**, **Dr. Sekar Viswanathan**, Vice Chancellor **Dr. V. S. Kanchana Bhaaskaran**; Pro-Vice Chancellor **Dr. Partha Sharathi Mallick**, and Registrar **Dr. Jayabarathi T**.

My whole-hearted thanks to Dean **Dr. Daphne Lopez,** School of Computer Science Engineering and Information Systems, Head, Department of Information Technology, **Dr. Arivuselvan K**, Information Technology Project Coordinator **Dr. Suganya P**, SCORE School Project Coordinator **Dr. Thandeeswaran R**, all faculty, staff and members working as limbs of our university for their continuous guidance throughout my course of study in unlimited ways

It is indeed a pleasure to thank my parents and friends who persuaded and encouraged me to take up and complete my project successfully. Last, but not least, I express my gratitude and appreciation to all those who have helped me directly or indirectly towards the successful completion of the project.

Place: Vellore

Date: **<Student Name>**

Executive Summary

Summary of the dissertation / internship for one page and not exceeding 200 words

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APPENDIX A .

Note: from Chapter 4 to Chapter 7 according to your **Project**, you are allowed to modify or to add the required chapters. Please discuss with your guide and add/modify the chapters accordingly.

List of Figures

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(In the chapters, figure caption should come below the figure and table caption should come above the table. Figure and table captions should be of font size 10.)

List of Tables

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List of Abbreviations

3GPP	Third Generation Partnership Project
2G	Second Generation
3G	Third Generation
4G	Fourth Generation
AWGN	Additive White Gaussian Noise

Symbols and Notations

δf	CFO
ε	NCFO

CHAPTER

INTRODUCTION

Artificial Intelligence (AI) refers to the development of computer systems that can perform tasks typically requiring human intelligence. These tasks range from recognizing speech, understanding natural language, making decisions, solving problems, to even learning from experience. AI encompasses a broad field of study, including machine learning, neural networks, robotics, natural language processing, and computer vision. By mimicking cognitive functions, AI aims to create systems capable of adapting to new information, making predictions, and optimizing processes.

As technology continues to evolve, AI has moved from theoretical concepts to practical applications that are transforming industries. From virtual assistants like Siri and Alexa to self-driving cars, healthcare diagnostics, and personalized recommendations, AI is reshaping how we live, work, and interact with the world around us. Understanding AI is crucial as it increasingly influences our daily lives, presents ethical challenges, and offers unprecedented opportunities for innovation and efficiency.

1.1 Objective

The primary objective of this AI project is to develop a [specific AI system/tool/application] that utilizes machine learning and data analysis techniques to [solve a particular problem or enhance a process]. The project aims to…

1.2 Motivation

An AI-based Traffic Management System aims to optimize the flow of vehicles in urban areas by leveraging machine learning algorithms and real-time data. Utilizing inputs from traffic cameras, sensors, and GPS data, the AI can predict traffic congestion, adjust traffic light timings dynamically, and suggest optimal routes for drivers. This reduces travel time, minimizes fuel consumption, and decreases carbon emissions.

By analyzing historical traffic patterns and integrating with public transportation systems, the AI can provide more accurate traffic forecasts and support city planners in making data-driven infrastructure decisions,

Page Specification:

Left Margin: 1.5 inch

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Font & Size:

Entire Dissertation - Font Name: Times New Roman

Font size: 12

Chapter Heading Font size: 14 and CAPS, BOLD

Sub Heading Font size: 12 and No CAPS

Paragraph:

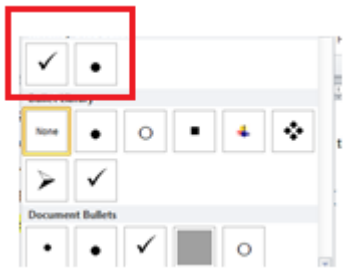
Paragraph Alignment: Justify

Paragraph line Space: 1.5 line space

Paragraph: In every chapter's first paragraph, first line without 1 tab space and continue, in next paragraph with 1 tab space till the end of the chapter.

*For reference check Chapter 1 formatting in this document

Bullet points: do not use bullet numbering, instead use bullet image, select



* for types, classifications , you can use bullet numbering (where it is required to mention in numbers)

Page Numbers:

- ✓ Use roman letters page numbering from cover page to symbols and Notations chapter.
- ✓ For Cover page do not make the page number visible.
- ✓ From page number ii , should be visible before the chapter 1
- ✓ Use numeric page numbers (1, 2,3,..) from **chapter 1. Introduction** till References sections
- ✓ For Appendices, No page numbers required
- ✓ Total Number of pages required: 70 and above (from chapter 1 to reference section)
- ✓ In Appendices: you can add any other sample codes, images, manuals, Readme.txt etc.,)

Figures:

- ✓ Centre the figure, Make it clear and visible, for better view go for portrait format. If the figure is huge in size then go for landscape format for image.
- ✓ In the chapters, figure caption should come below the figure.
- ✓ Figure Caption should be in the Title Case, Times New Roman, 10 pt, and it should be of the Format: **Fig. Chapter Number.Figure Number** Figure Caption
- ✓ In below example Figure , Second chapter. 4th image in the second chapter, title of the figure.

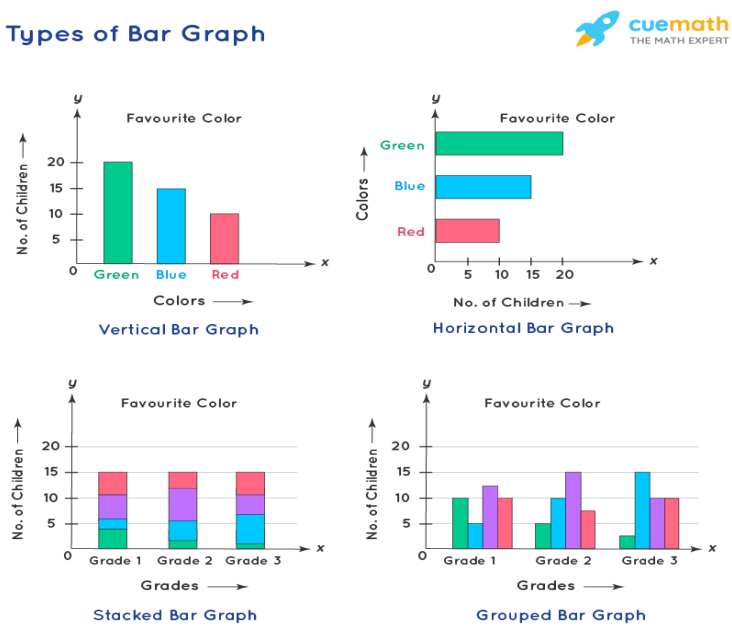


Fig. 2.4 Sample Bar Graph

Tables:

- ✓ For Tables: Title of the table should come in the top of the table as follows
- ✓ All the table caption should be in the Title case , Times New Roman, 10Point and it should be of the format: **Table.Chapter Number.Table number** Table Caption

Table 6.1 MI and Deep Learning

SAMPLE TABLE

Artificial Intelligence	Machine Learning	Deep Learning
AI stands for Artificial Intelligence, and is basically the study/process which enables machines to mimic human behaviour through particular algorithm.	ML stands for Machine Learning, and is the study that uses statistical methods enabling machines to improve with experience.	DL stands for Deep Learning, and is the study that makes use of Neural Networks(similar to neurons present in human brain) to imitate functionality just like a human brain.
AI is the broader family consisting of ML and DL as it's components.	ML is the subset of AI.	DL is the subset of ML.
AI is a computer algorithm which exhibits intelligence through decision making.	ML is an AI algorithm which allows system to learn from data.	DL is a ML algorithm that uses deep(more than one layer) neural networks to analyze data and provide output accordingly.

Equations:

Every equation should be used by relevant tools or symbols in the word editor.

Every equation should be cited as Eqn. 5.2 (Equation. fifth chapter.2 equation)

$$A^2+B^2=C \tag{5.2}$$

When you cite the above equation should cite as follows;

In the Eqn. 5.2 the variable A denotes the parameter of..

CHAPTER 8

SUMMARY

A brief summary of the dissertation / internship findings, expected results. Recommendations

CHAPTER 9

REFERENCES

SAMPLE REFERENCES

[1.]Domenico Bloisi, Luca Iocchi, Michele Fiorini, Giovanni Graziano, "Automatic Maritime Surveillance with Visual Target Detection", International Defense and Homeland Security Simulation Workshop (DHSS), September 2011.

[2.]Domenico Bloisi, Luca Iocchi, Michele Fiorini, Giovanni Graziano, "Camera Based Target Recognition for Maritime Awareness", Information Fusion (FUSION), 15th International Conference, July 2012.

[3.]Rodrigo Da Silva Moreira, Nelson Francisco Favilla Ebecken, Alexandre Soares Alves, Frédéric Livernet4, Aline Campillo-Navetti, "A Survey on Video Detection and Tracking of Maritime Vessels", IJRRAS 20, July 2014.

[4.]Sergiy Fefilatyev, "Algorithms for Visual Maritime Surveillance with Rapidly Moving Camera", Doctoral dissertation, University of South Florida, 2012.

Websites

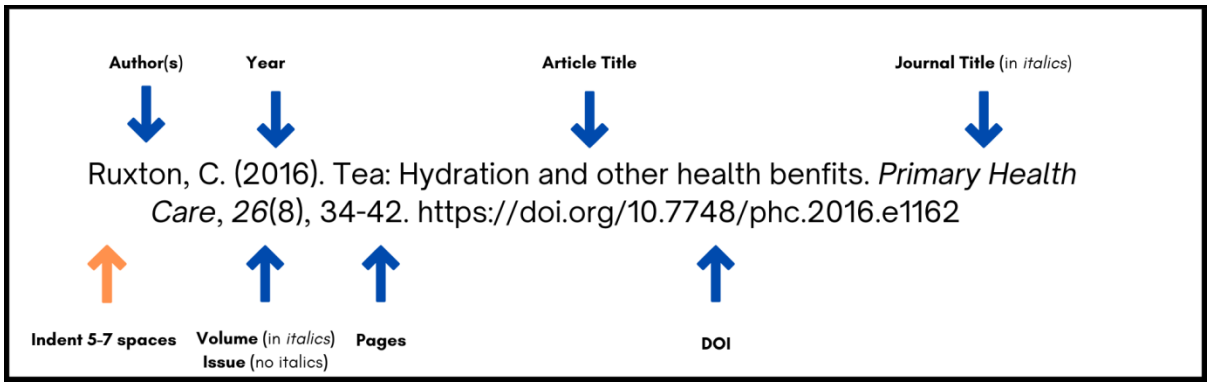
[5] Digi-key Corporation. http://www.digikey.com/

Data Sets

[6] Alldatasheet. http://www.alldatasheet.com/

Use APA Format for citation and reference section

Basic format to reference journal articles



Single Author format for APA Citation

INTEGRATING SOURCES (SUMMARIZED OR PARAPHRASED IDEAS):

	APA
Explanation	**Author's name:** Either within a signal phrase or in parentheses before the period at the end of the sentence. **Year:** Either within parentheses after the name that is used in a signal phrase or after the name and a comma within the parentheses before the period at the end of the sentence (name, year).
Example #1	Starr (2010) indicated that teachers' lack of personal understanding of technology could cause roadblocks to integrating technology into classrooms.
Example #2	Teachers' lack of personal understanding of technology can cause roadblocks to integrating technology into classrooms (Starr, 2010).

TWO AUTHORS:

	APA
Example #1	Merriman and Nicoletti (2008) suggested that US K–12 education must take on a structure that is globally acceptable.
Example #2	US K–12 education must take on a structure that is globally acceptable (Merriman & Nicoletti, 2008).

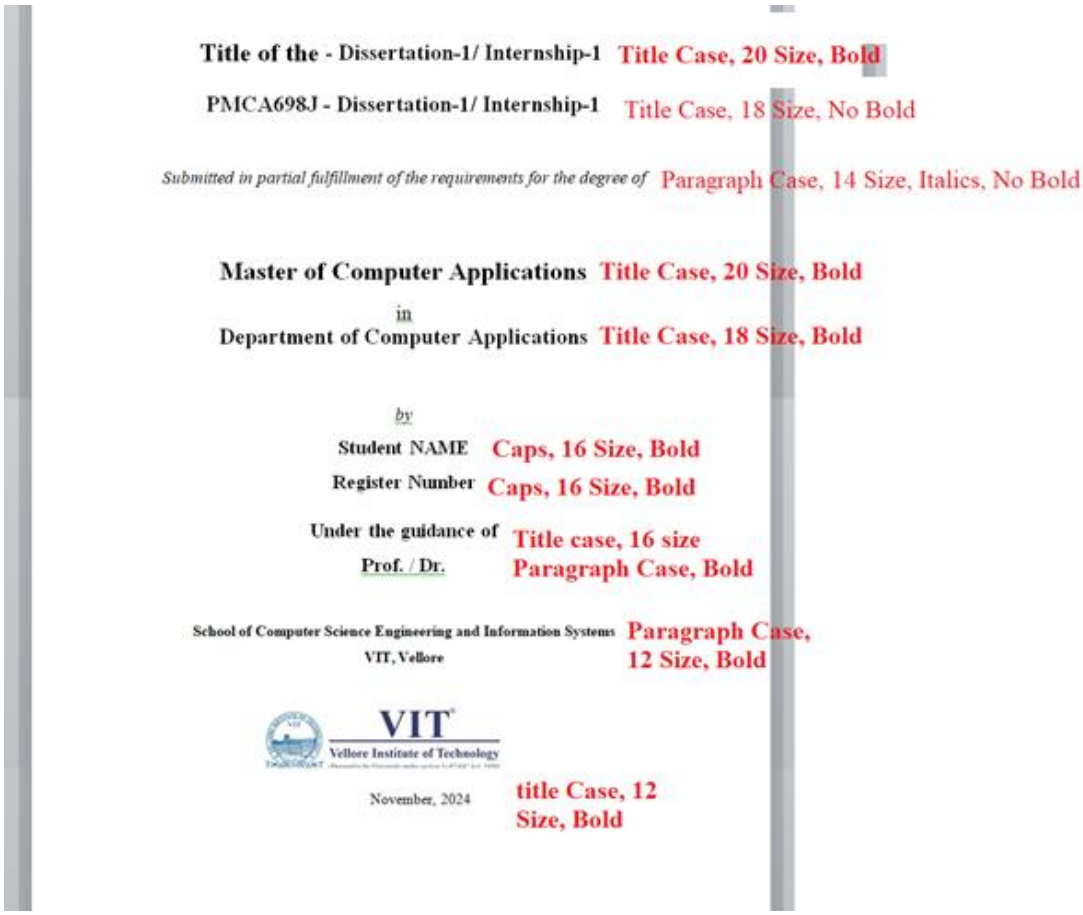
MULTIPLE AUTHORS:

	APA
Explanation	**Three to five Authors:** List all three authors at first reference (name, name, & name) and the first name plus "et al." for subsequent references (name et al.). **Six or more authors:** For all references, list the first name plus "et al." (name et al.).
Example #1	Borsheim, Merritt, and Reed (2008) suggested that teachers do not have to give up traditional curricula in order to integrate technology.
Example #2	In fact, it has been argued that technology has become part of education without a great deal of effort from teachers (Borsheim et al., 2008).

PERSONAL COMMUNICATION:

	APA
Example #1	M. Stanforth (personal communication, July 17, 2010) indicated she had been using a computer board in her classroom for three years and could not imagine giving it up.
Example #2	Many teachers are angry that they are being pushed to include technology because they like the way their classrooms work without it (S. Kennedy, personal e-mail, June 25, 2009).

APPENDIX A

- **Page numbers are not required from Appendix section**
- You can keep additional sample codes and sample screen shots of your Dissertation / Internship in appendix A; and inside the dissertation, wherever required, the codes, algorithms, images should be kept. In Appendix whatever you keep are additional only
- Don't keep the entire codes in the documents; use sample codes only, whichever is important.
- Screen shots of the Dissertation / Internship ; use sample images only, whichever is important

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